LAB REPORT

COURSE INFORMATION

- **Course Code:**
- **Course Title:**
- **Experiment No:**
- **Experiment Name:**

SUBMITTED TO

-
- **Department of**
- **East West University**

SUBMITTED BY

- **Student Name :**
- **ID :**
- **Section :**
- **Department :**

East West University

Date of submission:

Data Sheet: Diffraction grating experiment:

1 inch = 2.54 cm. ***1 nm (nanometer)= 10^{-7}cm = 10^{-9}m***

$$Vernier\ ConstAnt\ (V.C.) = \frac{Value\ of\ the\ smallest\ division\ in\ \ Main\ scale}{Total\ number\ of\ divisions\ \ in\ \ Vernier\ scale} = \ldots\ldots\ldots\ldots$$

Grating constant, ***N*** = the number of lines or rulings per centimeter of the grating surface

=…………………….

Number of order, ***n***	Description of the line	Reading for the **angle of Diffraction *θ***								λ = Wave length, $= \frac{Sin\theta}{nN}$ ***nm***
		Left Side			**Right Side**					
		Main scale reading (***S***) in degree	Vernier scale reading (***V***) $V = V.C.\times Div.$ in degree	Total reading (***L***) $L = S + V$ in degree	Main scale reading (***S***) in degree	Vernier scale reading (***V***) $V = V.C.\times Div.$ in degree	Total reading (***R***) $R = S + V$ in degree	$2\theta = L \sim R$ in degree	***θ*** in degree	
1st order	Violet									
	Indigo									
	Blue									
	Green									
	Yellow									
	Orange									
	Red									
2nd order	Violet									
	Indigo									
	Blue									
	Green									
	Yellow									
	Orange									
	Red									

Calculation: Wavelength, $\lambda = \dfrac{Sin\theta}{nN}$ = ...

	Violet	Indigo	Blue	Green	Yellow	Orange	Red
Standard value of Wavelength of different Color of **Sodium Light** (mainly **yellow color**)Source ***nm***	447	450	471	502	588	620	668
Standard value of Wavelength of different Color of **Mercury Light** (mainly **Violet color**)Source ***nm***	405	436	-	546	578	615	-
Experimental Value of wavelength in ***nm***	….	….	….	….	….	….	….

$$\text{Percentage Difference} = \frac{\text{Standard value} \sim \text{Experimental value}}{\text{Standard value}} \times 100\%$$

Percentage difference (Violet color) = ……….

Percentage difference (Indigo color) = ……….

Percentage difference (Blue color) = ……….

Percentage difference (Green color) = ……….

Percentage difference (Yellow color) = ……….

Percentage difference (Orange color) = ……….

Percentage difference (Red color) = ……….

Results:

	Violet	Indigo	Blue	Green	Yellow	Orange	Red
Wavelength in *nm*	….	….	….	….	….	….	….